

Darpan Radadiya

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SUMMARY

Data engineer and applied ML practitioner with hands-on experience in test automation, data quality validation, and model robustness verification across production pipelines. Built and maintained a 185-function pytest suite gating an ML pipeline's correctness, and designed data quality checks that prevented bad records from reaching production models. Strong Python and SQL foundation with exposure to PyTorch, Hugging Face Transformers, and ONNX Runtime model serving. MS in Analytics, Northeastern University, December 2026.

EDUCATION

Northeastern University

Master of Professional Studies in Analytics

December 2026 | GPA: 3.96/4.0

Dhirubhai Ambani University

Bachelor of Technology in Information and Communication Technology

Gujarat, India

June 2024 | CGPA: 7.45/10.0

SKILLS

- **AI/ML Testing & Validation:** Model output validation, robustness and edge-case testing, data quality assurance, regression testing, pytest-based test automation, pre-commit quality gates, retry/timeout and checkpoint-recovery validation, LLM output evaluation.
- **AI & Machine Learning:** PyTorch, scikit-learn, Hugging Face Transformers, ONNX Runtime model serving, Whisper speech-to-text, ArcFace/InsightFace embeddings, HDBSCAN clustering, OpenCV, LLM orchestration and evaluation (Gemini, OpenAI APIs), prompt engineering, feature engineering, time-series analysis.
- **Programming & Automation:** Python, SQL, R, JavaScript, C, C++; test automation and scripting, workflow automation, distributed task queues (Redis, RQ), job scheduling; Git, GitHub, pytest, Jupyter.
- **Data Engineering:** ETL/ELT pipeline development, data transformation, AWS Glue, Lambda, S3, QuickSight, PySpark, batch processing, REST API design with FastAPI, data quality validation rules.
- **Databases & Reporting:** PostgreSQL, MySQL, SQLite, Redis, Firestore; relational schema design, 3NF normalization, ERD documentation, query optimization, KPI dashboards (Tableau, Power BI, Streamlit, Excel).
- **Foundations:** Data structures and algorithms (600+ problems solved), operating systems, database management systems, object-oriented programming; Node.js, React.

EXPERIENCE

Data Analyst Intern, Wisaxis LLC

Boston, MA

Sep 2025 – Mar 2026

- Built reproducible Python and SQL analysis workflows over a 100K+ daily-record operational dataset with no prior scalable process, cutting time-to-insight by 30%.
- Applied LLM-based summarization and machine learning workflows to detect anomalies and inconsistent patterns in structured operational data, shortening issue-resolution cycles.
- Translated ambiguous business requirements into testable data specifications, and authored integration guides and best-practice documentation to standardize quality expectations across teams.

Data Analyst Intern, Clomotech Pvt. Ltd.

Gujarat, India

Jan 2024 – Jun 2024

- Built and maintained automated ETL pipelines on AWS Glue, Lambda, and S3 processing 100K+ operational records daily, replacing a legacy ingestion process that delayed downstream analytics.
- Developed Python preprocessing and data-cleaning scripts enforcing data quality validation rules before records reached machine learning models, reducing manual data-wrangling effort by 40%.
- Owned API-based ingestion troubleshooting and defect resolution, and built AWS QuickSight and Excel dashboards surfacing operational KPIs and early-warning quality signals for business stakeholders.

PROJECTS

Carbon Record Automation – Python, FastAPI, Redis, PyTorch, ONNX Runtime, Whisper

[GitHub](#)

- Architected a 270-module, 95K-line Python system that ingests feature-length video and automatically produces per-character vertical short-form reels, deployed as a FastAPI service with Redis and RQ workers under APScheduler orchestration.
- Built the machine learning pipeline end to end: Whisper speech-to-text, PySceneDetect shot segmentation, InsightFace ArcFace embeddings served through ONNX Runtime, and HDBSCAN clustering to resolve cast identities behind a human review gate – a robustness safeguard against low-confidence model output.
- Orchestrated multi-model LLM calls for cluster labeling and scene scoring with configurable timeouts and three-attempt retry logic, and added an on-disk embedding cache with per-stage checkpoints so multi-hour jobs resume instead of restarting – validated against failure and edge-case scenarios.
- Enforced correctness with **185 pytest functions across 24 test files** behind a pre-commit gate, covering functional, regression, and edge-case scenarios across the pipeline.

Healthcare Clinic ERP and Analytics Platform – SQL, Node.js, Express, React

[GitHub](#)

- Designed a 10-table clinical schema (3NF) covering patients, providers, encounters, ICD-10 diagnoses, invoices and payments, with documented ERD, loaded with 5,000 patients and 12,000 appointments to validate query performance at scale.
- Built provider-productivity, billing, and clinical-diagnosis reporting layers with KPI dashboards and a live SQL viewer tracing metrics to source tables; full CRUD REST API served to a React front end with CSV export.